

NBA Player Archetype Discovery

Unsupervised Learning · 2023–24 Season

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The Problem with Traditional Positions

Traditional 5-Position System

PG • SG • SF • PF • C

- Designed decades ago
- Fails to capture modern versatility
- Nikola Jokić listed as a Center but leads the league in assists
- Victor Wembanyama plays nothing like a traditional big man

Data-Driven Archetypes

K-Means • PCA • Hierarchical

- Let the statistics define the groups
- Captures the positionless evolution of modern NBA basketball
- Groupings based on actual on-court performance, not roster labels

Dataset & Feature Engineering

485

Players
After Filtering

19

Features
Used

2

Source
Files Merged

10+ mpg
& 20+ G

Minimum
Threshold

Scoring

PTS · FG · 3P · FT · FG% · 3P% · TS%

Playmaking

AST · TOV per game

Rebounding

ORB · DRB · TRB per game

Defense

STL · BLK per game

Advanced

PER · Win Shares · BPM · VORP

Methodology

01

Data Preparation

Filtered to 2023–24 season. Deduplicated traded players (kept TOT row). Merged Per Game + Advanced on player_id. Applied 10 mpg & 20 games minimum threshold.

02

Preprocessing

Applied StandardScaler to normalize all 19 features to zero mean and unit variance.

03

Optimal k Selection

Tested $k = 2-12$. Evaluated using Elbow Method (inertia) and Silhouette Analysis.

04

Clustering & Validation

K-Means ($n_init=20$) as primary method. Hierarchical Clustering (Ward linkage) as validation. Adjusted Rand Index = 0.583 confirms moderate agreement between methods.

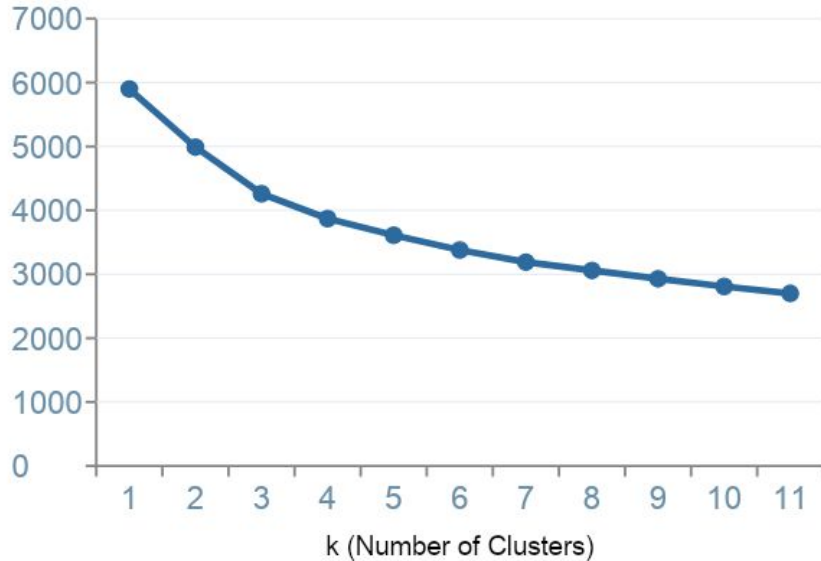
05

Visualization & Interpretation

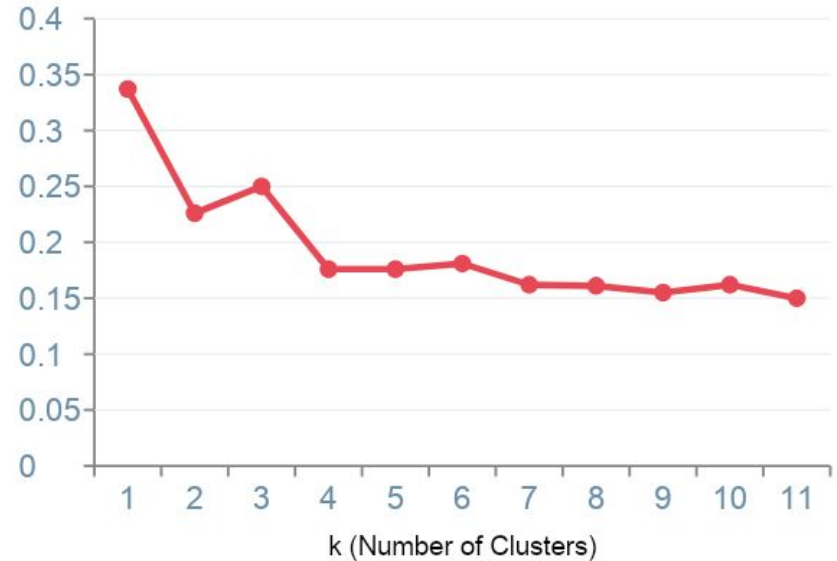
PCA reduced 19 features to 2D ($PC1+PC2 = 69.7\%$ of variance). Biplot revealed feature contributions. KNN similarity search validated archetype assignments against known player profiles.

Selecting the Optimal Number of Clusters

Elbow Method



Silhouette Analysis



Decision: $k = 4$ · Silhouette peaks at $k=2$ (0.337) but yields only 2 trivial groups. Secondary peak at $k=4$ (0.250) provides the best meaningful separation. Elbow curve confirms flattening after $k=4-5$.

The Four Player Archetypes

Cluster 0 · Paint Big

~60 players

High ORB (+1.63) · High BLK (+1.00) · High TRB (+1.10)

Traditional big men dominating the paint. Elite rebounders and shot-blockers with low 3-point activity (−1.02). Think Rudy Gobert, Clint Capela.

Cluster 2 · Bench / Limited Role Player

~214 players

All features negative · MP −0.80 · Low across the board

End-of-roster players with limited production across all metrics. Low minutes and low statistical output in every category.

Cluster 1 · Elite Star

~52 players

PTS +1.99 · AST +1.62 · PER +1.78 · VORP +2.10

Superstars producing at elite levels across every category. High usage, high minutes, top impact metrics. Jokić, Giannis, LeBron, Luka, Curry.

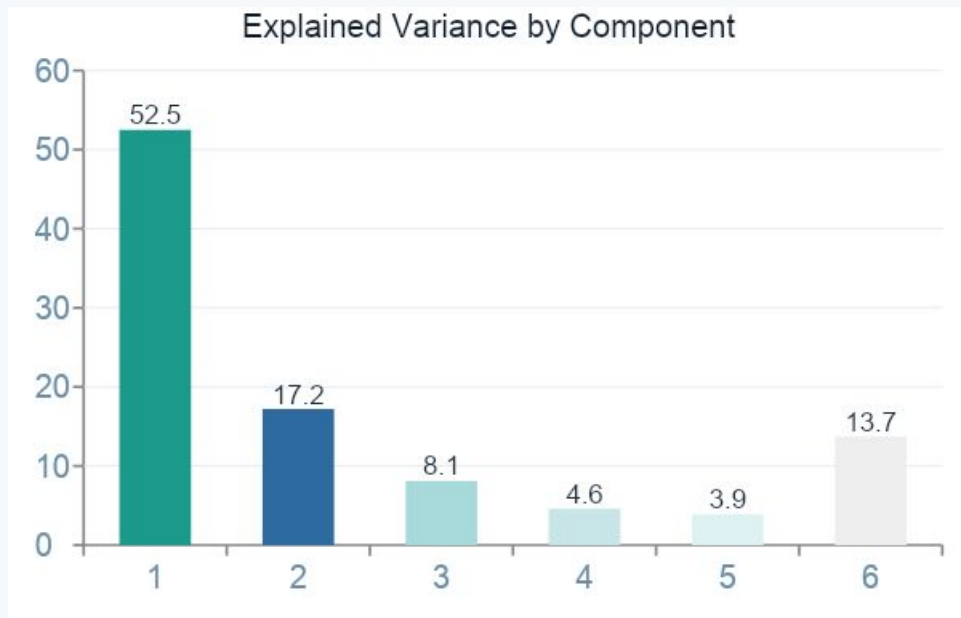
Cluster 3 · Perimeter Wing / 3-and-D

~143 players

3P +0.74 · STL +0.43 · PTS +0.49 · Low BLK/REB

Modern perimeter contributors who space the floor and defend. Below-average paint presence but strong 3-point output and solid steal rates.

PCA — Dimensionality Reduction



PC1 acts as an overall quality axis — clusters separate cleanly left-to-right from Bench to Wings to Bigs to Stars.

52.5%

PC1 Explained Variance

General overall production — scoring, efficiency, advanced metrics all load positively onto PC1.

17.2%

PC2 Explained Variance

Separates big-man profile (high rebounds/blocks) from perimeter profile (high 3-point rate).

69.7%

PC1 + PC2 Combined

Just two components capture nearly 70% of all statistical variance across 19 features.

3 PCs

To Reach 80% Threshold

The high compression confirms strong underlying structure in NBA player statistics.

Validation — Hierarchical Clustering

0.583

Adjusted
Rand Index

K-Means vs. Hierarchical
(1.0 = perfect · 0.0 = random)

K-Means \ HClust	H0	H1	H2	H3
Paint Big (K0)	5	4	47	4
Elite Star (K1)	0	45	0	7
Bench (K2)	190	0	1	23
Wing/3-D (K3)	25	0	12	106

- Strong agreement on Elite Stars (K1): 45 of 52 match exactly
- Strong agreement on Bench players (K2): 190 of 214 match exactly
- Overlap between Paint Bigs and Wings reflects real positional fluidity
- ARI of 0.583 is solid for real-world player data with genuine boundary cases

KNN — Statistical Similarity Lookup

Nikola Jokić

C · Cluster 1 · Elite Star

Giannis Antetokounmpo

30.4 pts · 11.5 reb · 6.5 ast

Domantas Sabonis

19.4 pts · 13.7 reb · 8.2 ast

Luka Dončić

33.9 pts · 9.2 reb · 9.8 ast

LeBron James

25.7 pts · 7.3 reb · 8.3 ast

Model ignores position labels — groups Jokić with elite multi-category dominators regardless of size.

Stephen Curry

PG · Cluster 1 · Elite Star

Tyrese Maxey

25.9 pts · 3.7 reb · 6.2 ast

Kyrie Irving

25.6 pts · 5.0 reb · 5.2 ast

Paul George

22.6 pts · 5.2 reb · 3.5 ast

Desmond Bane

23.7 pts · 4.4 reb · 5.5 ast

All comps are high-volume perimeter scorers with strong 3-point output — statistically accurate.

Victor Wembanyama

C · Cluster 1 · Elite Star

Chet Holmgren

16.5 pts · 7.9 reb · 2.3 blk

Anthony Davis

24.7 pts · 12.6 reb · 2.3 blk

Kristaps Porziņģis

20.1 pts · 7.2 reb · 1.9 blk

Myles Turner

17.1 pts · 6.9 reb · 1.9 blk

Modern stretch bigs with elite block rates. Myles Turner (Cluster 0) highlights Wemby's unique profile.

Key Findings

1

Positionless Basketball is Real

K-Means grouped Jokić, Giannis, and LeBron James together — three different traditional positions — because their multi-category statistical profiles are genuinely similar.

2

Four Archetypes Beat Five Positions

$k=4$ hit the best meaningful silhouette score (0.250). The data naturally collapses the 5-position system into four statistically coherent types that reflect how the modern game is played.

3

Strong Compression with PCA

PC1 alone explains 52.5% of variance (general quality axis). PC2 captures big-vs-perimeter. Just 2–3 components capture 80%+ of all statistical variation across 19 features.

4

Methods Agree on Clean Cases

ARI of 0.583 between K-Means and Hierarchical Clustering. Strong agreement on Elite Stars and Bench clusters. Disagreement concentrates on boundary players between Paint Bigs and Wings.

5

KNN Produces Intuitive Player Comps

Wembanyama's nearest neighbors: Chet Holmgren, Anthony Davis, Porzingis — all modern stretch bigs with elite block rates. The model builds useful comparison tools as a byproduct of clustering.

Multi-Season Clusters Analysis

4

Seasons

25

Features

4

Archetypes

85%

PCA Variance

15+ mpg
& 30+ G

Methodology

01

Data Cleaning

Merged per-game & advanced stats.
Removed traded-player duplicates.
Min. 30 GP · 15 min/game.

02

Feature Engineering

25 stats scaled with StandardScaler.
PCA retaining 85% variance reduces noise before clustering.

03

K-Means (k=4)

k selected via Elbow + Silhouette.
k=4 chosen to match single-season analysis.
n_init=20 for stable centroids.

04

Validation

Archetype Transition Matrix shows season-to-season label stability.
Adjusted Rand Score gives a single summary metric.

The Four Player types (Multi Season)

Cluster 0 · Paint Big

237 players

PTS 10.5 · DRB 4.85 · BLK 0.95 · 3P only 0.48 · TS% 62%

Traditional bigs dominating the paint. Elite rebounders and shot-blockers with very low 3-point activity. PER 17.8, VORP 1.07. Think Gobert, Capela. ~72% C/PF.

Cluster 1 · Elite Star

131 players

PTS 24.8 · AST 6.1 · PER 22.9 · VORP 4.04 · TS% 60%

Superstars producing at elite levels across every category. High usage, high minutes, top impact metrics. Jokić, Giannis, LeBron, Luka, Curry. All positions.

Cluster 2 · Bench / Limited Role Player

680 players

PTS 8.3 · AST 1.9 · PER 11.5 · VORP 0.12 · Low across board

Largest cluster. End-of-rotation players with limited production in every category. Low minutes, low impact metrics. All positions represented.

Cluster 3 · Perimeter Wing / 3-and-D

343 players

PTS 16.6 · 3P 2.05 · STL 0.98 · PER 15.6 · Low BLK/REB

Modern perimeter contributors who space the floor and defend. Strong 3-point output and steal rates, below-average paint presence. VORP 1.23.

Archetype Stability & Position Alignment

508 unique players · 152 in all 4 seasons · avg stay rate 79.8%

Archetype Transition Matrix

% staying in same archetype season-to-season



Avg. stay rate: 79.8% , strong diagonal confirms archetypes capture real player types

Archetypes vs Traditional Positions

% of each archetype made up by each position



Only Paint Big aligns with a position (72% C), all others span positions equally

Key Findings



Archetypes Are Highly Stable

79.8%

avg stay rate

Average stay rate of 79.8% across all archetype transitions. Bench/Role Players (84%) and Elite Stars (82%) show the strongest retention superstar status and limited roles are persistent.



Position Labels Are Obsolete

72%

Paint Bigs are C

Only Paint Big aligns with a traditional position (72% Centers). Elite Stars are 34% PG, 25% PF, modern stars defy positional categories. Perimeter Wings are evenly split across SG/PG/SF.



Elite Stars Are a Rare Tier

9.4%

of seasons

Only 131 of 1,391 player-seasons (9.4%) qualify as Elite Stars. With PTS 24.8, AST 6.1 and VORP 4.04, this cluster is well-separated, Jokić, Giannis, LeBron, Luka, Curry.



Bench Role is the Largest Cluster

49%

of all seasons

680 of 1,391 seasons (49%) fall in Bench/Limited Role. With PER 11.5 and VORP 0.12, these are genuine contributors but replaceable — and they come from all 5 positions evenly.

Conclusion & Next Steps

What We Found

- Traditional positions fail to capture modern NBA versatility
- Data-driven clustering reveals 4 meaningful archetypes
- Silhouette score of 0.250 at $k=4$ confirms cluster quality
- PCA compresses 19 features into 2D with 69.7% variance retained
- KNN similarity search produces intuitive, accurate player comps
- Archetypes are stable over time 79.8% average stay rate across 4 seasons confirms clusters capture real player types, not noise.
- Stricter filter (30 GP, 15 min) strengthens consistency across seasons compared to the single-season 20 GP threshold
- Bench/Role Player is the dominant archetype — 49% of all qualified player-seasons fall there

Next Steps

- Build interactive player comparison dashboard
- Apply to lineup optimization and draft evaluation
- Explore DBSCAN for density-based clustering

Thank You!

